

```

1  #include <Wire.h>
2  #include <TFT_eSPI.h>
3  #include <CST816S.h>
4
5  #include "USB.h"
6  #include "USBHIDKeyboard.h"
7  USBHIDKeyboard Keyboard;
8  //
9  // #define TFT_MOSI 15 // In some display driver board, it might be written as "SDA" and so c
10 // #define TFT_SCLK 14
11 // #define TFT_CS 5 // Chip select control pin
12 // #define TFT_DC 27 // Data Command control pin
13 // #define TFT_RST 33 // Reset pin (could connect to Arduino RESET pin)
14 // #define TFT_BL 23
15 TFT_eSPI tft = TFT_eSPI(); // Use default SPI pins
16 CST816S touch(5, 6, 10, 7); // sda, scl, rst, irq
17
18 // Keyboard parameters
19 int keySizeX = 40; // Width of each key
20 int keySizeY = 30; // Height of each key
21 int keyboardWidth = 6 * keySizeX;
22 int keyboardHeight = 4 * keySizeY;
23
24 String typedText = ""; // String to store typed letters
25
26 void setup() {
27   Serial.begin(115200);
28   Keyboard.begin();
29   USB.begin();
30   // Initialize the display
31   tft.init();
32   tft.setRotation(0); // Set the rotation as needed

```

Fig. 6: Code setting for the TFT_eSPI library and I2C pins

```

24 String typedText = ""; // String to store typed letters
25
26 void setup() {
27   Serial.begin(115200);
28   Keyboard.begin();
29   USB.begin();
30   // Initialize the display
31   tft.init();
32   tft.setRotation(0); // Set the rotation as needed
33
34   // Initialize the touch controller
35   touch.begin();
36
37   // Set the background color
38   tft.fillScreen(TFT_BLACK);
39
40   // Draw the keyboard higher up on the display
41   int keyboardX = (tft.width() - keyboardWidth) / 2;
42   int keyboardY = (tft.height() - keyboardHeight) / 3; // Adjusted to move the keyboard up
43
44   // Draw red boundary lines
45   tft.drawFastHLine(keyboardX, keyboardY - 1, keyboardWidth, TFT_RED);
46   tft.drawFastHLine(keyboardX, keyboardY + keyboardHeight, keyboardWidth, TFT_RED);
47   tft.drawFastVLine(keyboardX - 1, keyboardY, keyboardHeight, TFT_RED);
48   tft.drawFastVLine(keyboardX + keyboardWidth, keyboardY, keyboardHeight, TFT_RED);
49
50   for (int row = 0; row < 4; row++) {
51     for (int col = 0; col < 6; col++) {
52       char key = 'A' + (row * 6) + col;
53       drawKey(keyboardX + col * keySizeX, keyboardY + row * keySizeY, key);
54     }
55   }

```

Fig. 7: Code snippet for keyboard

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“
Branding is
what people
say about
you when
you are not
in the room.
”



-Jeff Bezos

